

How these sheets work

One sheet per competency, 7 sheets for this video. Each gives you two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before the video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing, some a grid, some three panels. The division is yours.
4. Words go inside the drawing, as labels and inside little boxes with arrows. A sentence running the width of the box does not count.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a grid or a sequence, so whichever way you think, there is a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set it up on a tablet, and write the prompt letter you chose at the top. Hand ruled counts exactly the same.

This is anatomy

Every prompt asks what a structure is, where it sits, and what it is next to. Function belongs to physiology and is not on these sheets.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

1

COMP

WEEK 4 · BONE HISTOLOGY

Cartilage types and structure

You should be able to: Identify the three cartilage types by fiber content and location and label chondrocytes, lacunae, matrix, and perichondrium.

PICK ONE AND DO THAT ONE

☐ **A** LABELED DRAWING, THREE PANELS

Divide the box into three panels, one per cartilage type. In each, draw the tissue as it looks under the microscope and label chondrocytes, lacunae, matrix, and the perichondrium where it exists. Draw the fibers so the three panels look different from each other.

Then: Under each panel, write one place in the body where you would find it.

☐ **B** SORTING GRID

Draw a grid with hyaline, elastic, and fibrocartilage down the side. Across the top: the fiber that dominates, how the matrix looks on a slide, one location, and whether a perichondrium is present. Fill every cell with a word or a tiny sketch, not a sentence.

Then: Circle the one cell in the grid that lets you tell fibrocartilage from the other two at a glance.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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2

WEEK 4 · BONE HISTOLOGY

Cartilage growth modes

COMP

You should be able to: Distinguish appositional from interstitial cartilage growth by where each begins and the direction of expansion.

PICK ONE AND DO THAT ONE

☐ **A SPLIT DRAWING**

Draw a line down the middle. On the left, a piece of cartilage growing appositionally: show the perichondrium and arrows for where the new matrix is added. On the right, the same piece growing interstitially: show chondrocytes dividing inside the matrix and arrows for the direction it expands.

Then: Under each half, one line naming where that growth starts.

☐ **B BEFORE AND AFTER SEQUENCE**

Draw a small block of cartilage three times in a row: before, during, and after growth. Do the row twice, once for appositional and once for interstitial, so the two rows sit one above the other and the difference in where the new tissue appears is obvious.

Then: Mark on each row the single structure that is the source of the new cartilage.

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3

COMP

WEEK 4 · BONE HISTOLOGY

Bone classification by shape

You should be able to: Classify a bone as long, short, flat, irregular, or sesamoid and give an example of each.

PICK ONE AND DO THAT ONE

☐ **A SKELETON MAP**

Draw a rough skeleton outline. Pick one bone for each of the five shapes, draw it in place, and label it with the shape it belongs to. Use a different mark for each shape so the five classes stand out.

Then: Add one more bone for the shape you find hardest to remember.

☐ **B FIVE-COLUMN SORT**

Draw five columns headed long, short, flat, irregular, sesamoid. Under each heading, sketch the outline that defines the shape, then list two example bones. For sesamoid, draw the tendon the bone sits inside.

Then: Put a star beside the one example you think students most often put in the wrong column, and write which column they wrongly choose.

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COMP

WEEK 4 · BONE HISTOLOGY

Gross anatomy of a long bone

You should be able to: Label the diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and marrow regions on a long bone.

PICK ONE AND DO THAT ONE

☐ **A LABELED SECTION**

Draw a long bone cut down its length so the inside shows. Label diaphysis, both epiphyses, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, and the epiphyseal line. Shade compact bone and spongy bone differently and write where red and yellow marrow sit.

Then: Draw the nutrient foramen and the perforating fibers, and label both.

☐ **B OUTSIDE-IN LADDER**

Draw a ladder that walks from the outside of a long bone to the center of its shaft: periosteum, compact bone, endosteum, medullary cavity, marrow. Then draw a second ladder for the end of the bone: articular cartilage, compact shell, spongy bone, red marrow. Beside each rung, a tiny sketch of what that layer looks like.

Then: Mark on the second ladder where the epiphyseal line sits and what it used to be.

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5

COMP

WEEK 4 · BONE HISTOLOGY

Compact and spongy bone microanatomy

You should be able to: Label the parts of an osteon and contrast compact bone with the trabeculae of spongy bone by structure and location.

PICK ONE AND DO THAT ONE

☐ **A LABELED DRAWING, ONE OSTEON**

Draw one osteon large in cross section. Central canal in the middle, concentric lamellae as rings, lacunae between the rings with an osteocyte in each, canaliculi as fine lines linking them. Add a perforating canal crossing to a neighboring osteon, and fit interstitial lamellae into the gaps between osteons.

Then: In the corner, draw a small patch of spongy bone and label a trabecula and the marrow space beside it.

☐ **B SIDE-BY-SIDE COMPARISON**

Draw a line down the middle. Left: compact bone as it looks on a slide. Right: spongy bone on a slide. Under each, write its structural unit, what fills its spaces, and where in a long bone it sits.

Then: Circle the one structure that appears on the left and never on the right, and name it. That is your slide giveaway.

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COMP

WEEK 4 · BONE HISTOLOGY

The four bone cells

You should be able to: Identify the osteogenic cell, osteoblast, osteocyte, and osteoclast and state the role and origin of each.

PICK ONE AND DO THAT ONE

- ☐ A

LINEAGE LADDER PLUS ONE OUTSIDER
Draw three rungs for the lineage, osteogenic cell to osteoblast to osteocyte, with a small drawing of each cell on its rung and where it sits in the bone. Draw the osteoclast off to the side, large and multinucleated, with its own arrow showing it comes from a blood cell line.

Then: Beside each cell write its job in three words or fewer.
- ☐ B

SORTING TABLE WITH SKETCHES
Draw a table with the four cells down the side. Across the top: what it looks like, where in the bone you find it, what it does, and where it comes from. Draw the cell in the first column instead of describing it.

Then: Put a box around the one row where the origin is different from the other three.

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7

COMP

WEEK 4 · BONE HISTOLOGY

Ossification and growth plate zones

You should be able to: Contrast intramembranous and endochondral ossification and list the five growth plate zones in order from epiphysis to diaphysis.

PICK ONE AND DO THAT ONE

☐ **A LABELED GROWTH PLATE**

Draw the growth plate as a tall column with the epiphysis at the top and the diaphysis at the bottom. Draw the five zones in order and label each: resting cartilage, proliferation, hypertrophy, calcification, ossification. Sketch what the chondrocytes look like in each zone so the zones are visibly different.

Then: Beside the column, write which zone lies against the epiphysis and which against the diaphysis.

☐ **B TWO-ROUTE SPLIT**

Draw a line down the middle. Left: intramembranous ossification, the tissue it starts from, the bones it builds, and a small drawing of a skull bone. Right: endochondral ossification, the hyaline cartilage model it starts from, the bones it builds, and a small drawing of a long bone.

Then: Along the bottom, list the five growth plate zones in order as a single arrow from epiphysis to diaphysis.

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